OR4023 (Optimization) Homework 6

Name: _____ Student ID: _____

Question 1. Let f be a differentiable function that is being minimized with an optimization algorithm that uses an exact line search. Let $\mathbf{p}_k$ be the search direction at $\mathbf{x}_k$, show that $\nabla f(\mathbf{x}_{k+1})\mathbf{p}_k = 0$.

Question 2. Prove that if $f(x) = \frac{1}{2}\sum_{i=1}^{m} r_i(x)^2$, then

$$\nabla^2 f(x) = \sum_{i=1}^{m} \left[\nabla r_i(x)\nabla r_i(x)^T + r_i(x)\nabla^2 r_i(x)\right].$$

Question 3. Given data points $(t_i, y_i)_{i=1}^m$, determine the least squares fit by the function

$$y = x_1 + x_2 \cos t + x_3 \sin t$$

by finding the normal equation. Then find the least squares solution for the following data:

t_i	0	1	2	3	4
y_i	1.4	1.6	1.2	0.6	0.3

and draw the data points and the fitted curve.

Question 4. Consider the least squares problem of fitting the data points

t_i	1	2	4	5	8
y_i	3.3	4.3	7.2	9	20

by the model $y = x_1 e^{x_2 t}$. Conduct two iterations of

(a) the classical Newton's method

(b) the Gauss Newton

Question 5. Consider the constrained problem

$$\begin{aligned} \text{minimize} \quad & f(\mathbf{x}) = x_1^2 + x_1^2 x_3^2 + 2x_1 x_2 + x_2^4 + 8x_2 \\ \text{subject to} \quad & 2x_1 + 5x_2 + x_3 = 3 \end{aligned}$$

Determine which of the following points are stationary points: (i) $(0, 0, 2)^T$; (ii) $(0, 0, 3)^T$; (iii) $(1, 0, 1)^T$.

Question 6. Use the method of Lagrangian multipliers to solve

$$\begin{aligned} \text{minimize} \quad & f(\mathbf{x}) = x_1^3 + x_2^3 \\ \text{subject to} \quad & 2x_1 + x_2 = 1 \end{aligned}$$